Hybrid Retrieval for Long Document Question Answering

Abstract

Retrieval systems that answer questions about a single long document usually rank passages with BM25 alone. We introduce ConvoQA, a dataset of 1842 human questions written over 96 technical PDFs, and a hybrid ranker that combines Okapi BM25 with an idf-weighted coverage gate. On the dev split the hybrid ranker reaches an F1 of 0.82 against 0.74 for BM25 alone, while adding no model download to the serving path.

1 Introduction

Spoken assistants raise the bar for retrieval: a wrong passage is heard as a confident lie, and a slow answer is heard as a broken device. We therefore measure retrieval by the evidence it puts in front of the answerer rather than by ranking metrics alone.

Prior work on open-domain QA optimises recall at fifty passages. Our setting is the opposite: four passages, one document, and one second of wall clock budget.

2 The ConvoQA Dataset

Questions were written by six annotators against public laboratory manuals, policy handbooks and vendor datasheets. Each question is paired with the page and the exact sentence that answers it. Disagreement between annotators was resolved by a third reader, and 41 questions were discarded because no passage in the source PDF answered them.

The split is 80 percent training, 20 percent evaluation, with no document shared between them.

3 Method

Passages are formed by a line-aware chunker that keeps a section heading with the lines under it and targets sixty words. Ranking is Okapi BM25 with k1 of 1.5 and b of 0.75 over two fields, body and heading, where the heading score is multiplied by 1.4.

A candidate is accepted only when the idf-weighted coverage of the query terms in the best passage exceeds 0.34. Coverage uses only terms that occur somewhere in the document, so a long question is not penalised for vocabulary the document never had.

4 Results

Table 1	Ranker	Precision at 4	Answer F1	Median latency
BM25	0.69	0.74	4 ms	
BM25 plus heading field	0.75	0.78	4 ms	
Hybrid with coverage gate	0.81	0.82	5 ms	
Cross encoder reranker	0.86	0.85	214 ms	

The coverage gate is responsible for most of the gain at no measurable cost, and the reranker buys four points of F1 for two orders of magnitude of latency.

5 Ablation

Removing the heading field costs 0.04 F1. Replacing line-aware chunks with fixed windows of 500 characters costs 0.03 F1 and breaks code quotation, because a single statement is no longer a retrievable unit. Removing ordinal expansion costs 0.02 F1 on questions that name a section.

6 Conclusion and Future Work

A single long document can be answered well with a hand-written ranker. Future work covers spoken barge-in, where the assistant must stop mid-sentence, and multi-document retrieval over a shared corpus. We release ConvoQA and the chunker under the MIT licence.

References

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